

2023-24 |

Bharat6G
Alliance

ANNUAL REPORT

BHARAT 6G ALLIANCE



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EXECUTIVE SUMMARY

On March 22, 2023, the Department of Telecommunications (DoT) released the Bharat 6G Vision document, outlining India's ambitious goals for 6G technology. The Bharat 6G Vision is to "Design, develop, and deploy 6G network technologies that provide ubiquitous, intelligent, and secure connectivity for a high-quality living experience globally." The document emphasizes positioning India as a global leader in intellectual property, developing affordable 6G telecom solutions, and deploying these technologies to enhance India's capabilities by 2030, ultimately improving the quality of life for people worldwide.

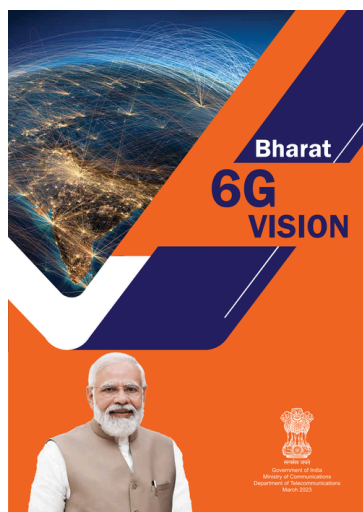
The Bharat 6G Vision is founded on the principles of Affordability, Sustainability, and Ubiquity. It aligns with the national vision of Atmanirbhar Bharat, aiming to empower every Indian to become self-reliant. Additionally, it seeks to establish India as a key player in the global market for advanced telecom technologies and solutions that are affordable and contribute to the greater good. The timing of this vision aligns with India's celebration of 75 years of independence, known as Aazadi Ka Amrit Kaal, and envisions India as a leading contributor in the design, development, and deployment of 6G technology by 2030.

The Bharat 6G Vision document also emphasizes the creation of a robust ecosystem for research, innovation, and collaboration among academia, industry, and government. By fostering partnerships across sectors, India aims to accelerate the development of cutting-edge 6G technologies, strengthen its position in the global telecommunications landscape, and contribute to the establishment of global standards. This collaborative approach ensures the alignment of India's technological advancements with the evolving demands of global connectivity and digital transformation.

Moreover, the vision document highlights the critical role of 6G technologies in addressing pressing global challenges, including climate change, energy efficiency, and digital inclusivity. With a focus on sustainability, the deployment of 6G is expected to enable smart solutions for urban planning, healthcare, education, and disaster management, thereby promoting equitable growth. By leveraging advanced technologies such as artificial intelligence, machine learning, and quantum computing, India aims to deliver transformative solutions that benefit not only its citizens but also the global community.



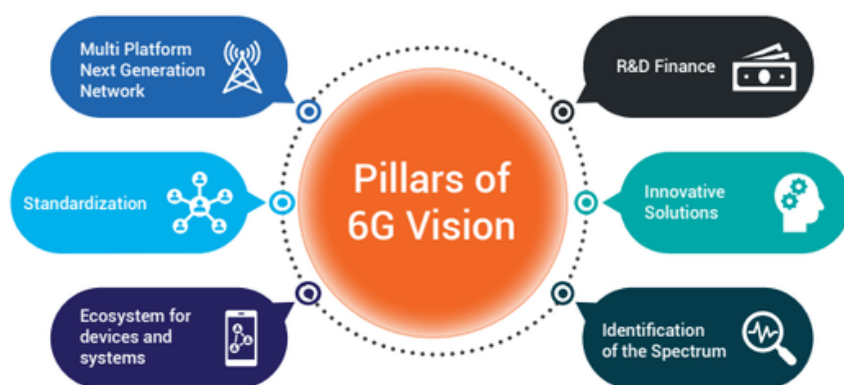
BHARAT 6G VISION



The Department of Telecommunications (DoT) constituted a Technology Innovation Group on 6G (TIG-6G) on 1st November 2021 with members from various Ministries/Departments, research and development institutions, academia, standardization bodies, Telecom Service Providers, and industry to develop Vision, Mission, and Goals for 6G and develop a roadmap and action plans for 6G in India. The TIG-6G in turn constituted six Task Forces with industry, academia, R&D institutions and Government as members on Multi-Disciplinary Innovative Solutions, Multiplatform Next Generation Networks, Spectrum for Next Generation Requirements, Devices, International Standards Contribution and Funding Research and Development.

The Six Task Forces under TIG-6G deliberated on various aspects of emerging telecom technologies and platforms in the next decade. They focused on innovations that leverage these new technologies to deliver solutions, the device ecosystem that will support these innovations, spectrum management that will enable the ongoing and oncoming explosion in wireless communications, the need to contribute our innovations to global standards and ensure interoperability, and requirement of adequate financing to carry out the Bharat 6G Mission.

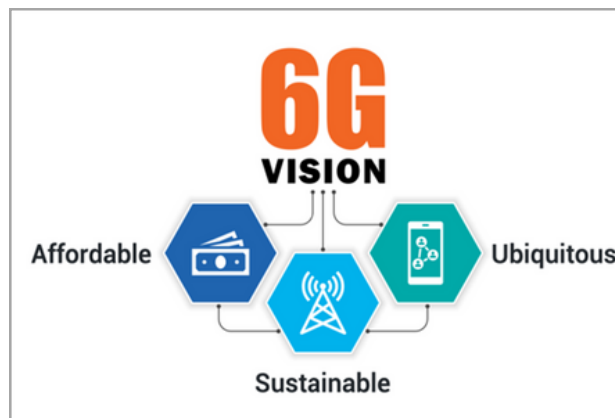
Based on the reports of the Six Task Forces, a Bharat 6G Vision Document has been prepared by the Technology Innovation Group on 6G.



Six task forces formed under the Technology Innovations Group



BHARAT 6G VISION STATEMENT



6G Vision is based on principles of Affordability, Sustainability, and Ubiquity. The Vision for 6G is fully aligned with the national Vision of Atmanirbhar Bharat and will seek to empower every Indian to become Atmanirbhar (self-reliant) in their lives. At the same time, it ensures that India takes its rightful place in the world as a leading supplier of advanced telecom technologies and solutions that are affordable and contribute to the global good. The 6G Vision is thus timed just right for India's Aazadi ka Amrit Kaal.

India gearing up for 6G era: Modi

FINAL TOUCHES. Prime Minister releases roadmap and action plan to launch the network in the country

S Ronendra Singh
New Delhi

Prime Minister Narendra Modi, on Wednesday, launched the Bharat 6G Vision Document to develop a roadmap and action plan for 6G network in the country.

Addressing a gathering after inaugurating the International Telecommunication Union (ITU) Area Office and Innovation Centre, Modi said India rolled out 5G mobile technology at a fast pace.

"Within six months of the rollout of 5G technology, we are talking about 6G today. 5G services have been expanded to 125 cities within 120 days of the launch," said Modi.

TECH EXPORTER

The vision document was prepared by the Technology



MASTERMINDS. (from left) S Jaishankar, Minister of External Affairs; Prime Minister Narendra Modi; Ashwini Vaishnaw, Union Minister for Communications, Electronics & Information Technology; and Doreen Bogdan-Martin, International Telecommunication Union Secretary-General, at the inauguration of ITU Area Office and Innovation Centre, in New Delhi, on Wednesday. (VISHAL NARAIN)

Innovation Group on 6G (TIG-6G), which was constituted in November 2021 with members from various Ministries/Departments, research and development institutions, academia, standardisation bodies and telecom service providers.

"Before 4G, India was only a user of telecom technology, but now India is moving fast to become a big

exporter of telecom technology," he said. Modi also announced that the country would soon set up 100 new 5G labs. "These labs will help in developing 5G applications, in line with the unique needs of India," he said.

Stressing that India's telecom and digital model is smooth, secure, transparent and trusted, Modi called

this decade the 'techade'. The number of Internet users in the country increased from 25 crore in 2014 to 85 crore, with the majority in rural areas.

Modi also added the government and private sector together have laid over 25-lakh km of optical fibre in the last nine years and more than five lakh common service centres

provide online services to rural areas. The opening of the ITU area office follows India signing a Host Country Agreement in March 2022 with ITU.

ECONOMIC TIES

The new area office, fully funded by India, will serve India, Nepal, Bhutan, Bangladesh, Sri Lanka, Maldives, Afghanistan and Iran, enhancing coordination among the nations and fostering mutually beneficial economic cooperation in the region, said a statement.

NEW APP LAUNCHED

The Prime Minister also launched the 'call before u dig' (CBUD) mobile app that links agencies that need to dig up roads or carry out other excavation work to ensure that underground structures or cables are not damaged.



Bharat 6G: Modi launches prep for next-gen telecom

Also inaugurates International Telecommunication Union Area Office

SUBHAYAN CHAKRABORTY
New Delhi, 22 March

Prime Minister Narendra Modi on Wednesday unveiled India's 6G mission, which visualises a focused initiative for the next-generation tech in India, and the official 6G testbed project to allow research on new network technologies.

The government has launched the Bharat 6G project, and appointed an apex council to oversee it. India is the second-largest telecom market globally with 1.2 billion digital subscribers.

Modi said the past nine years witnessed an unprecedented digital leap, whereby India's digital economy grew 2.5 times faster than the national economy. Modi also launched the 'Call Before You Dig (CBuD)' app to facilitate coordination between excavation agencies and underground utility owners to prevent damage to utilities due to digging.

"Over this period, the number of broadband users rose to 800 million, from 60 million, and the number of internet connections rose to 850 million from 250 million. The government and private sector together have laid over 2.5 million km of optical fibre," he said. Modi stressed India is the most connected democracy in the world, where 70 million e-authentications are done every day, and 8 billion UPI transactions are made every month. "India has sent upwards of ₹28 lakh crore directly to its citizens through direct benefit transfers," he said.

6G supply chains

In November 2021, the gov-



“WITHIN SIX MONTHS OF THE 5G ROLL-OUT, TODAY WE ARE TALKING ABOUT 6G. THIS SHOWS THE CONFIDENCE OF INDIA. TODAY WE HAVE BROUGHT THE VISION DOCUMENT TO THE FORE. THIS WILL BECOME A BIG BASE FOR THE ROLL-OUT OF 6G”

6G: VISION & MISSION

- Mission to run in two phases till 2030
- To make India 6G R&D, manufacturing hub
- At 1Tbps, 6G to be 100 times faster than 5G
- Network to be built upon 5G infrastructure; be more reliable and affordable
- Suitable spectrum to be identified
- Hyper personalised networks, drone communications, and holographical on cards

ernment formed a 22-member innovation group headed by Telecom Secretary K Rajaraman to create a vision for the 6G road map in India.

Opening up 6G airwaves is part of India's efforts to attain a key foothold in the global supply chain for emerging technology, such as satellite and terrestrial com-

munication gear and components.

The new 6G testbed is being co-developed by a consortium of Indian Institutes of Technology (IITs), including Guwahati and Madras.

The project will provide an R&D platform to start-ups, researchers, industry and other broadband wireless

applications in India like e-Governance, smart cities, rural Broadband or other Digital India initiatives under Atmanirbhar Bharat.

New ITU office

The prime minister also inaugurated the first Area Office & Innovation Centre of the International Telecommunications Union (ITU) in India. ITU is the United Nations' specialised agency for information and communication technologies (ICT). The ITU's next World Telecommunication Standardisation Assembly will also be held in India next year.

The new office will help ITU coordinate with India and other countries in the region, said ITU's secretary-general Doreen Bogdan-Martin.

Modi unveils Bharat 6G vision document

NEW DELHI, MARCH 22

Prime Minister Narendra Modi on Wednesday unveiled a 6G vision document and said initiatives for 6G within six months of the 5G rollout showed India's confidence.

The 'Bharat 6G' vision document unveiled by PM Modi states 6G will offer speed up to 1 terabit per second, which is 1,000 times more than the top speed of 5G.

The document has been prepared by Technology Innovation Group on 6G (TIG-6G) constituted in November 2021. Its members are from ministries, departments, research and development institutions, academia, standardisation bodies, telecom service providers and industry.

The PM also inaugurated the International Telecommunication Union (ITU) Area Office and Innovation Centre, which has been set up on the Centre for Development of Telematics (C-DoT) campus. It will be fully

Says speed to be 1,000 times more than 5G



Prime Minister Narendra Modi at the inauguration of ITU Area Office and Innovation Centre in New Delhi on Wednesday. PTI

funded by India. PM Modi said the ITU office in India would help create the right environment for 6G in the country. "From being a mere consumer of telecom technology, India is now moving fast to become a big exporter of technology," he said.

"Within six months of the 5G rollout, we are today talking about 6G. This shows India's confidence. The vision document will become a big base for the rollout of 6G. India will set up 100 new 5G labs in the coming days. These labs will help develop 5G applications

as per the unique needs of India," he said. The 5G mobile technology have expanded to 125 cities within 120 days of the launch. "Before 4G, India was only a user of telecom technology, but now we are moving fast to become a big exporter in the sector," the PM said. Digital inclusion has benefited digital payments, direct benefit transfer, JanDhan, Aadhaar and the rapid rollout of broadband services, among other things PM Modi said.

The number of Internet users has increased from 25 crore in 2014 to 85 crore. Both government and private sectors have jointly laid out 25 lakh km of optical fibre in the country, which includes connectivity to about two lakh gram panchayats. The ITU is the United Nations' specialised agency for information and communication technologies, which coordinates with all member countries to create global standards for telecom technologies. — TNS

BHARAT 6G MISSION

The research and start-up ecosystem will be tapped to bring innovations and new ideas to the table. Adequate provisions for financial support will be made available. The Mission will be completed in two phases:

Phase 1 from 2023-2025

- 2023-24: Explorative ideas, Risky pathways, Research projects to build IPs for entering into Standards Arena, File IPs with potential for 6G.
- 2024-25: Proof-of-concept tests, Test & Measurement platforms for 6G work, Pre-standardization activities.

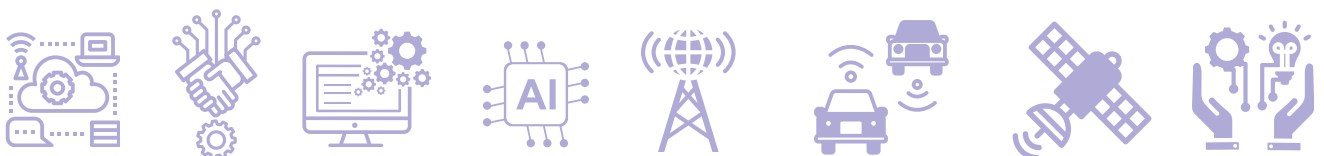
Phase 2 from 2026-2030

- Develop ideas and concepts that show promise and potential for acceptance by the global peer community,
- Establish their use cases and benefits, and
- Create IP and test-beds leading to commercialization.
- Addressing of regulatory and ethical considerations
- Public Engagement and Awareness

India has taken a significant leap toward 6G technology by unveiling the Bharat 6G Vision document and initiating the establishment of 6G test beds to foster innovation. With projections suggesting that India will become the world's largest wireless market by 2030, the nation's proactive involvement in 6G research and development positions it as a potential key player in shaping global standards. As the International Telecommunication Union (ITU) rolls out the IMT-2030 framework for 6G, India's early engagement in this technological evolution could play a pivotal role in defining critical aspects such as technology development, spectrum policies, and practical use cases.

BHARAT 6G MISSION - OBJECTIVES

1. Facilitate and finance Research and Development, design and development of 6G technologies by Indian Startups/Companies/Research bodies/Universities;
2. Enable India to become a leading global supplier of IP, products and solutions of affordable 6G telecom solutions;
3. Deploy 6G technologies to act as a powerful force multiplier for India@2030;
4. Enable inclusive and significant enhancement in the quality of living experience of citizens in India and across the world.

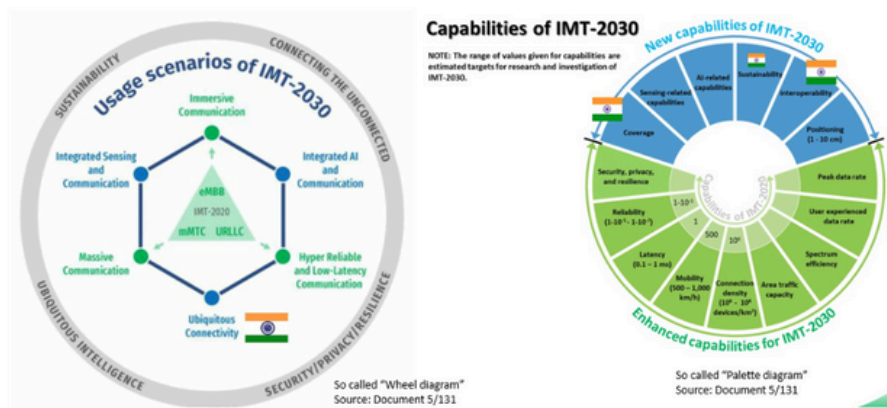


IMT 2030 FRAMEWORK

In July 2023, the Radiocommunication Sector of the International Telecommunication Union (ITU-R) achieved a consensus on the global vision for IMT-2030 (6G), marking a pivotal step in advancing 6G technology. The IMT-2030 Framework outlines 15 key capabilities for 6G, including nine enhancements to existing 5G features such as security and resilience, latency, mobility, connection density, peak data rate, and spectrum efficiency. The usage scenarios of IMT-2030 will build upon those established by IMT-2020 while also enabling entirely new use cases powered by 6G capabilities, such as AI integration and advanced sensing—areas beyond the design scope of earlier IMT generations.

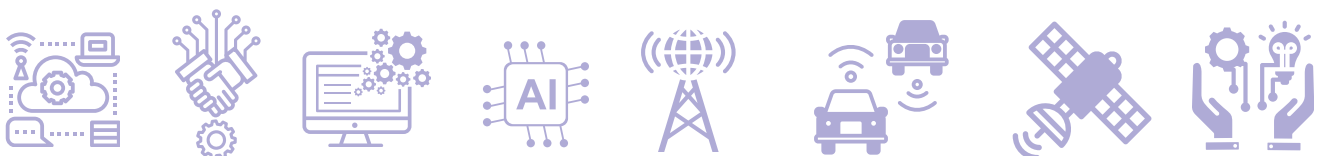
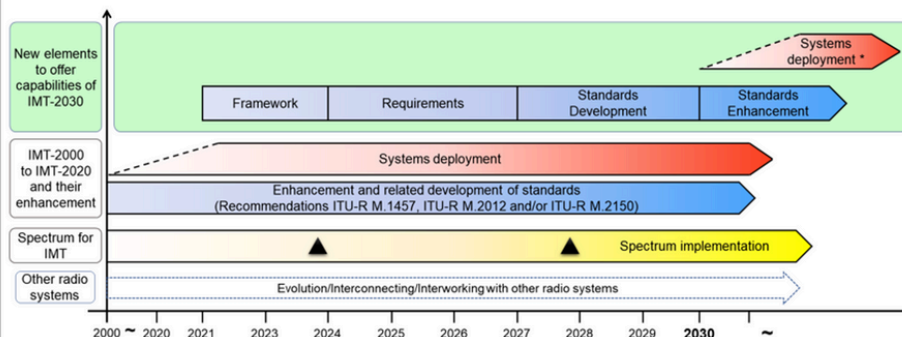
Under the Government's leadership, India has already prioritized Ubiquitous Connectivity, Comprehensive Coverage, Sustainability, and Interoperability as key elements aligned with its 6G Vision.

IMT-VISION 2030 Usage & Key Capabilities



Relationship and Timelines

- Roadmap for technology/standard development, deployment and spectrum
- In addition, enhancement of existing IMTs and relationship with other radio systems

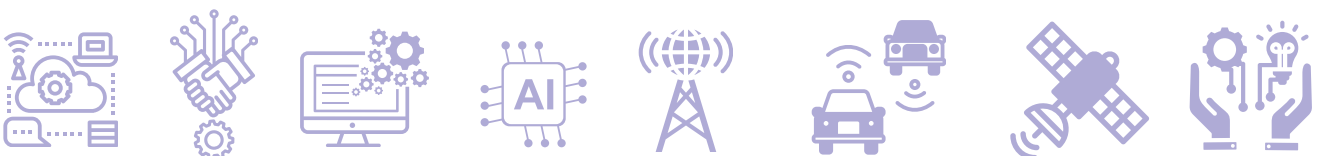


FORMATION OF BHARAT 6G ALLIANCE

In support of Bharat 6G Vision, the Bharat 6G Alliance (B6GA) was launched on July 3, 2023, by the Honourable Minister of Communications, Shri Ashwini Vaishnaw. The B6GA serves as a collaborative platform, bringing together public and private enterprises, academia, research institutions, and standards development organizations. The initiative aims to foster innovation in 6G network technologies, with a focus on providing intelligent and secure global connectivity. By aligning with India's Atmanirbhar Bharat vision, B6GA seeks to position India as a leader in the telecom sector, ensuring the affordability and sustainability of advanced technologies.



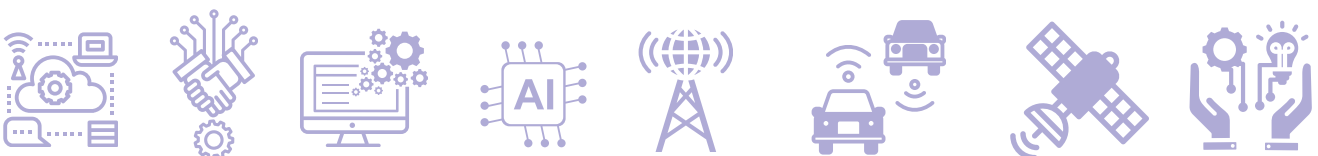
Launch of Bharat 6G Alliance by Honourable Minister of Communications, Shri Ashwini Vaishnaw, and Honourable Minister of State for Communications Shri Devusinh Chauhan on 3rd July 2023.



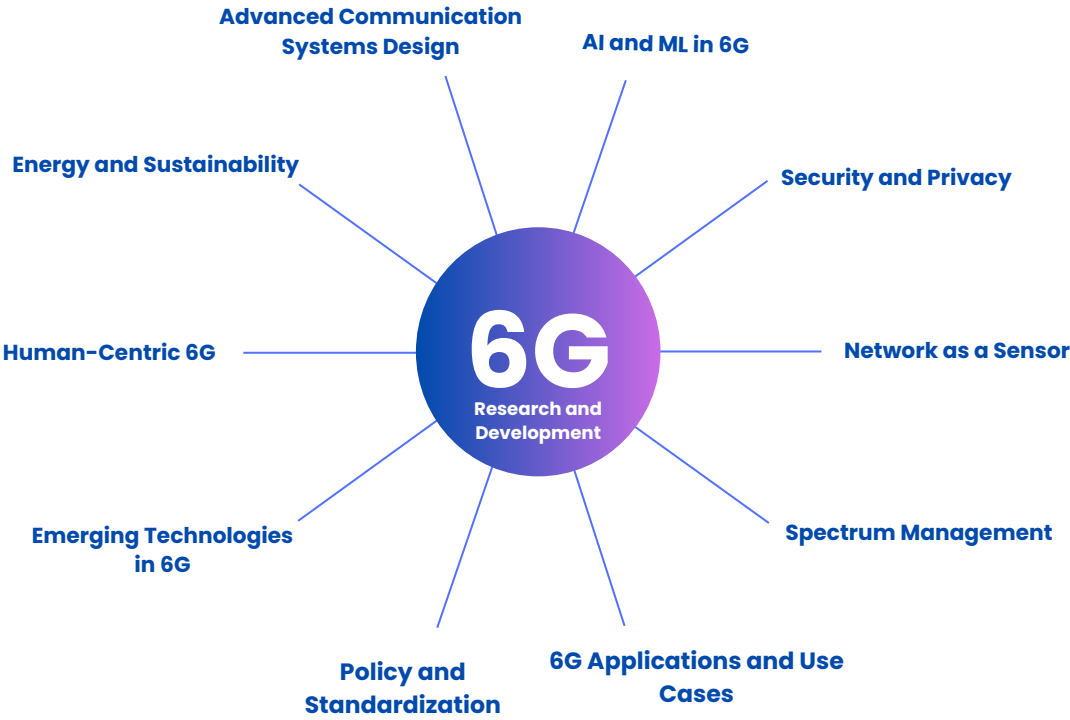
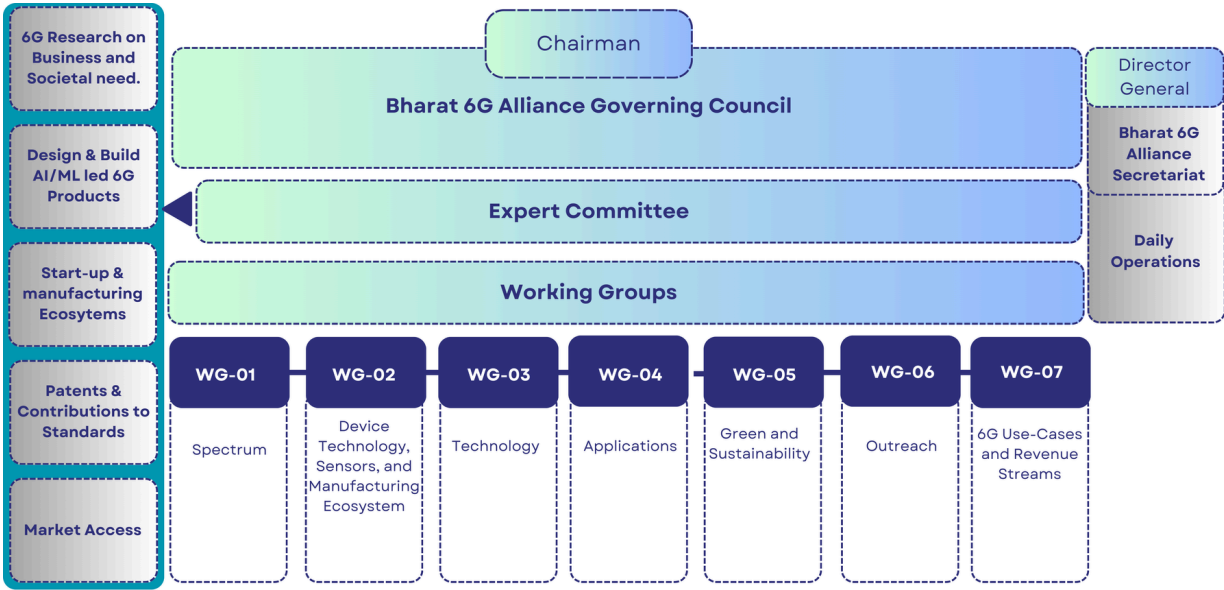
OBJECTIVES OF BHARAT 6G ALLIANCE

B6GA is an industry-led body, facilitated by the government, that provides a collaborative platform to various technology stakeholders comprising Public/Private companies, Academia, Research Institutions, and Standard Development Organizations.

- Promote impactful open research to advance 6G technologies
- Understand business and societal needs 5G advanced and 6G technology
- Pursue pre-standardization efforts to establish foundational frameworks for 6G
- Identify priority research areas that are crucial for 6G development
- Facilitate proof-of-concept prototypes and early market interventions led by startups
- Build coalitions with global 6G technology alliances to foster international collaboration
- Ensure access to 6G test beds and chipsets for testing and development
- Shape the future of global wireless communication technology by positioning India as a leader
- Establish India as a leading global supplier of intellectual property, products, and affordable 6G telecom solutions by 2030
- Aim to establish India as a global leader in affordable 5G and 6G technologies and deploy these technologies by 2030
- Support and energize Indian participation in standard development organizations
- Contribution to 6G and other future technology-related global standards, deployments, products, operations and services
- To bring together academic institutions, domestic industries, research and development entities, test labs, Government entities, telecommunication service providers, technology and innovation start-ups etc. as well as the Government of India, to help create synergies for a balanced growth of the technology and innovations ecosystem in the country for the benefit of the common people
- Support Make in India initiatives, bridging the gap between R&D and commercialization, and facilitate early market access for Indian 6G products

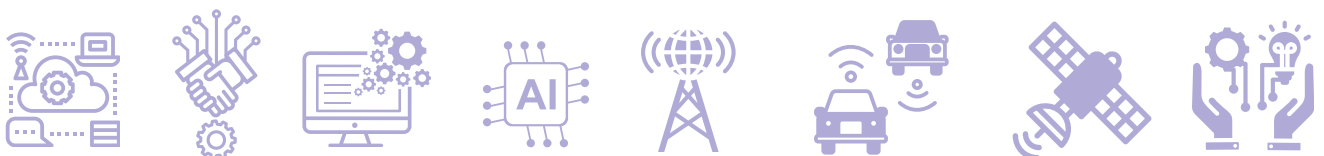


BHARAT 6G ALLIANCE ORGANISATION STRUCTURE



FOUNDING MEMBERS

S. No	Name & Designation
1	Mr N G Subramaniam (Chief Operating Officer, Tata Consultancy Services)
2	Prof Bhaskar Ramamurthi (Professor, Indian Institute Of Technology Madras)
3	Prof Abhay Karandikar (Director, Indian Institute of Technology Kanpur)
4	Mr Randeep Singh Sekhon (Chief Technology Officer Airtel, Bharti Airtel)
5	Mr Aayush Bhatnagar (Senior Vice President and Chief Technology Officer, Reliance Jio)
6	Mr Jagbir Singh (Chief Technology Officer, Vodafone Idea)
7	Dr Rajkumar Upadhyay (Chief Executive Officer, C DoT)
8	Dr. P Hanumantha Rao (Director General, Sameer)
9	Prof Kiran Kuchi (Founder & Director, Wisig Networks Private Limited)
10	Mr Ramu T S (Founder & Director, Lekha Wireless Solutions Private Limited)
11	Dr. Kumar N. Sivarajan (Chief Technology Officer, Tejas Networks Limited)
12	Mr Himamshu Khasnis (Chief Executive Officer, Signalchip Innovations Private Limited)
13	Mr Puneet Agarwal (Chief Executive Officer, VVDN)
14	Mr Raghuveer B.K. (Chief Executive Officer, Nivetti Systems Private Limited)
15	Mr Jagdish Mitra (Chief Strategy Officer, Tech Mahindra)
16	Mr Rajesh Kumar Pathak (Director, TCOE India)



MEMBERSHIP CATEGORY

Bharat 6G Alliance has five categories:

1. Corporate Members

(i) Domestic companies/entities incorporated and registered and having Headquarters in India, who are designing and/or manufacturing products, offering services, developing software/applications or involved in the testing of technology equipment (including start-ups, venture capital funds etc.) and Government and Government institutions.

(ii) Academia and Research & Development entities having Headquarters in India; and

(iii) Telecom Service Providers licensed to operate by the Government of India. Corporate Members shall have the power to elect the Governing Council and to participate in the various Expert Committees/Working Groups that may be constituted by the Society.

2. Affiliate Members

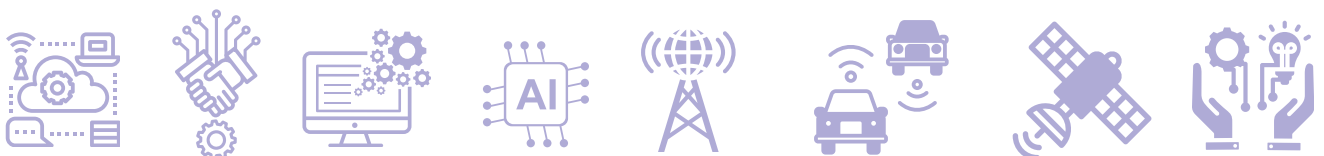
Foreign companies/entities incorporated and registered in India, who are designing and/or manufacturing products, offering services, research and development, developing software/applications or involved in the testing of technology equipment. Affiliate Members may participate in Working Groups for technical contribution and shall be part of the General Body. However, they shall not have voting rights in the election of Governing Council or any other Committee/Working Group.

3. Associate Members

Foreign companies/entities not registered in India and other associations registered in India, which share a common vision for the enhancement of technology and innovations, shall be eligible for admission as Associate Members. Associate Members may participate in Working Groups for technical contributions but shall not be part of the General Body.

4. Expert Members

Expert Members can be individuals who are self-employed or employed in an organization who shall be observers or Members with limited privileges. They will not form part of the General Body of Society. They shall be invited by the Chairperson for Expert Committees/Working Groups. Individuals identified by the Governing Council to participate in Expert Committees/Working Groups (WG) as Expert Member. The Expert Members should be experts in a specific area whose expertise an Expert Committee/WG is keen to leverage. The individual Expert Member must get the concurrence of his organization before the individual is invited. The tenure of the Expert Member shall be specified while inviting him/her to join the Expert Committee/WG.

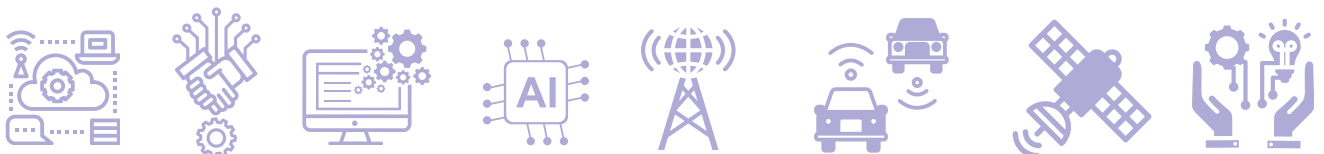


5. Guest Members

Guest Members can be from organizations who shall be observers with limited privileges. They will not form part of the General Body of Society or of GC. Based on GC approval, they will be invited by the Chairperson. In case of organizations which wish to explore the need for becoming a Member before taking up Corporate or Affiliate Membership, the Guest Membership shall be provided only for six months and shall not be renewed for three years unless the organization becomes a Corporate or Affiliate Member during that period.

GOVERNING COUNCIL

Chairman	Mr. N. G. Subramaniam (Chief Operating Officer, Tata Consultancy Services)
Members	Prof Bhaskar Ramamurthi (Professor, Indian Institute Of Technology, Madras)
	Prof Ajit Kumar Chaturvedi (Professor, Indian Institute Of Technology Kanpur)
	Mr. Randeep Singh, (Chief Technology Officer Airtel, Bharti Airtel)
	Mr. Aayush Bhatnagar (Senior Vice President and Chief Technology Officer, Reliance Jio)
	Mr. Jagbir Singh (Chief Technology Officer, Vodafone Idea)
	Dr Rajkumar Upadhyay (Chief Executive Officer, C DoT)
	Dr. P Hanumantha Rao (Director General, Sameer)
	Prof Kiran Kuchi (Founder & Director, Wisig Networks Private Limited)
	Mr. Ramu T S (Founder & Director, Lekha Wireless Solutions Private Limited)
	Dr. Kumar N. Sivarajan (Chief Technology Officer, Tejas Networks Limited)
	Mr. Himamshu Khasnis (Chief Executive Officer, Signalchip Innovations Private Limited)
	Mr. Puneet Agarwal (Chief Executive Officer, VVDN)
	Mr Raghuveer (Chief Executive Officer, Nivetti Systems Private Limited)
	Mr Jagdish Mitra, (Chief Strategy Officer, Tech Mahindra)
	Mr Rajesh Kumar Pathak (Director, TCOE India)
	Shri A Robert Jerard Ravi (Deputy Director General(SRI), DOT)



MEETINGS OF GOVERNING COUNCIL

1st Governing Council Meeting held on 24th May 2023

- (i) The First meeting of Governing Council and General Body was held on 24 May 2023 to consider and accept the Memorandum of Association of B6GA for submission to Department of Telecommunications (DOT).
- (ii) Mr. Rajesh Kumar Pathak, Director TCOE India was nominated as Member Secretary and Dr. Rajkumar Upadhyay, CEO CDoT was nominated as Member Treasurer of the Society for taking all necessary actions for the formation of the Society.

2nd Governing Council Meeting held on 26th August 2023

- (i) The 2nd meeting of GC was held on 26th August 2023 in which the GC resolved on Opening of bank account, its signatories and authority for GST, TAN/PAN etc.

The GC also decided on the Form for membership and fees for the different categories of membership as given in Table.

MEMBERSHIP FEE

S.No.	Membership Fee Category	Annual Fee (in Rs.)
1.	Corporate Members (India Headquartered industry and academia)	
	i) Upto Annual Turnover <= INR 10 Crore (Startup)	10,000
	ii) INR 10 Crore > Annual Turnover <= INR 50 Crore	50,000
	iii) INR 50 Crore > Annual Turnover <= INR 100 Crore	1,00,000
	iv) INR Above 100 Crore	5,00,000
	v) Academic Institutions, Not-for-Profit R&D Organizations, Government Department/Statutory or Autonomous bodies/Organizations set up by the Government or Statutory or Autonomous bodies	1,00,000
2.	Affiliate Members (Foreign companies incorporated and registered in India)	5,00,000
3.	Associate (Indian Associations)	1,00,000
4.	Associate (Foreign companies not registered in India)	USD 10,000
5.	Guest/Expert Members on Invitations	NIL

3rd Governing Council meeting held on 5th February 2024

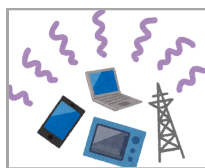
The 3rd Governing Council meeting was held on 5th February 2024 and following decisions were taken:

(i) Appointment of Auditor for Financial Year 2023-2024 and FY 2024-2025.

- Prem P Lamba appointed as Auditor.

(ii) Finalisation of following Working Groups (WGs).

1. Spectrum
2. Device Technology, Sensors and Manufacturing ecosystem
3. Technology
4. Applications
5. Green and Sustainability
6. Outreach, and
7. 6G Use-cases and Revenue Stream



Spectrum



**Device Technology,
Sensors, &
Manufacturing
Ecosystem**



Technology



Applications



Green & Sustainability



Outreach



**6G Use-Cases and
Revenue Streams**



(iii) The Governing Council finalised the maximum number of GC members in different verticals and Government Nominees on the GC as under:

TYPE OF MEMBERS IN GOVERNING COUNCIL

S.No.	Type of Vertical	Maximum Number of Members
1.	Telecom Service Providers (TSPs) (UASL)	03
2.	Other TSPs- (Other non UASL DoT licensees)	02
3.	Telecom Network Equipment	03
4.	Terminal Equipment Manufacturers: Manufacturers of Mobile Device/IoTs/Sensors/CPE/ End User Devices	02
5.	Startups (As defined by DPIIT)	02
6.	Academic Institutions	02
7.	Information Technology/Application Industry Applications/ Solutions Development/ Application Software / Service Platform Development/ Digital communication Software Development	02
8.	R&D organisations	02
9.	Government Department/ Statutory or Autonomous bodies/organisations set up by the Government or Statutory or autonomous bodies	02
10.	Semiconductor/Any other entity in ICT enabled product manufacturing/ service/ solution provider domain (not covered elsewhere)	01
11.	Bharat 6G Mission Head DoT, WPC, TEC (one each) Nominees	03
12.	MeitY, DST, Space, Defence, MHA and Railways (One each) Nominees	06
13.	DG	01
14.	Total	31



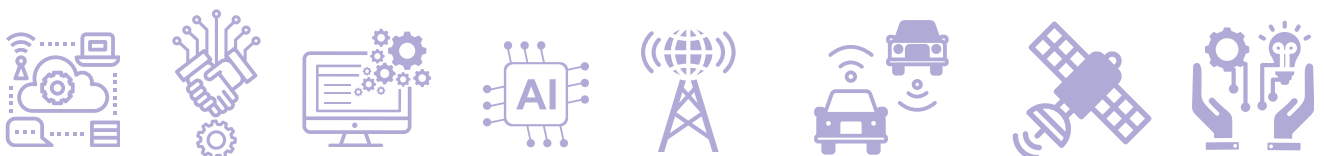
(iv) Decision on Number of Vice Chairs in Governing Councils, Expert Groups and Working Groups

The GC approved the maximum number of Vice chairs in GC, Expert Groups and Working Groups .

Maximum number of Vice Chairs in GC	02
Maximum Number of Vice Chairs in Expert	03
Maximum Number of Vice Chairs in Working Groups	03

1st Annual General Meeting (AGM) held on 5th February 2024

The AGM discussed the agenda items and endorsed the decision of Governing Council related to membership fee, Members from different verticals, Government Nominees in Governing Council and number of Vice Chairs position in Governing Council, Expert Groups and Working Groups. It also approved the budget for financial year 2023-2024 and financial Year 2024-2025.



MEMORANDUM OF UNDERSTANDING WITH GLOBAL ALLIANCES

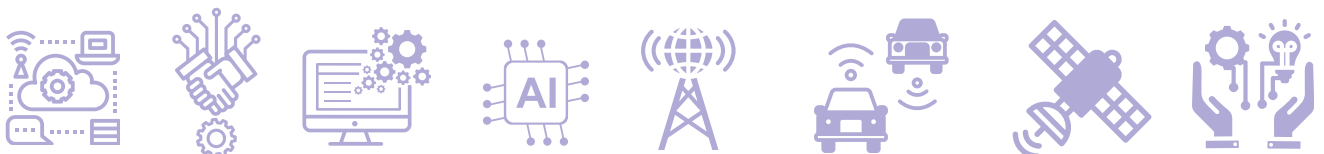


Bharat 6G Alliance signed MoUs with following global alliances to foster collaborations on 6G and related technologies:

1. ALLIANCE FOR TELECOMMUNICATIONS INDUSTRY SOLUTIONS (ATIS), THE NEXT G ALLIANCE(NGA)



The Next G Alliance, led by the Alliance for Telecommunications Industry Solutions (ATIS), is a pivotal initiative aiming to secure North American leadership in wireless technology, including 6G. Comprised of over 150 top global ICT companies, the alliance focuses on the full lifecycle of technology development, from research and standardization to market readiness and commercialization. Key priorities include network reliability, 5G, AI, cybersecurity, and IoT, among others. ATIS, accredited by ANSI, also partners with major international bodies like 3GPP and ITU, highlighting its role in advancing North American innovation and establishing strategic industry priorities.



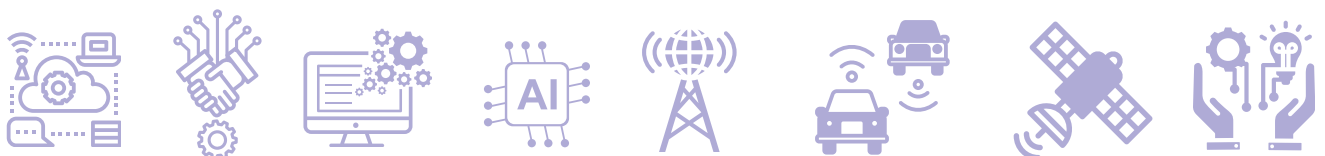
Bharat 6G Alliance inks pact with Next G Alliance to explore 6G wireless technology for telecom sector, supply chains



Alliance for Telecommunications Industry Solutions (ATIS)'s Next G [Alliance](#) and the Bharat 6G Alliance announced on September 9 they have signed a memorandum of understanding (MoU) to explore collaboration opportunities on 6G wireless technologies. The work will encompass aligning research and development priorities that support a common 6G vision and creating secure and trusted telecommunications as well as resilient supply chains, according to the statement.

The Bharat 6G Alliance entered into a Memorandum of Understanding with the Alliance for Telecommunications Industry Solutions (ATIS), specifically its Next G Alliance (NGA), on 31st August 2023 to foster a collaborative relationship. The collaboration is in line with the vision outlined in the Joint Statement by the United States and India on June 23, 2023. Both countries are dedicated to developing secure and trustworthy telecommunications systems, ensuring resilient supply chains, and promoting global digital inclusion, with a special emphasis on the advancement of 6G technologies. A key component of the collaboration involves supporting the development of open and interoperable wireless technologies.

Both alliances are committed to a full lifecycle approach encompassing research, design, prototyping, development, proof of concept testing, IPR creation, field testing, security, certification, and manufacturing. This holistic approach ensures the readiness and competitiveness of 6G technologies in the global market.



2. 6G SMART NETWORKS AND SERVICES INDUSTRY ASSOCIATION (6G-IA)



The 6G Smart Networks and Services Industry Association (6G IA) is an international non-profit organization based in Brussels, Belgium, dedicated to advancing 6G technologies. It focuses on collaboration in 5G and 6G research, pre-standardization, regulatory issues, and future frequency bands. The association aims to shape industrial use cases for Beyond 5G (B5G) and 6G, promoting a future with AI-powered, human- and machine-centric networks. 6G IA is committed to creating scalable, secure, and inclusive digital infrastructure, aligning with societal values and supporting Europe's technological sovereignty.

India Joins Forces With Europe in 6G Technology Push



India's Bharat 6G Alliance has recently signed a Memorandum of Understanding (MoU) with its European counterpart, the 6G Smart Networks and Services Industry Association (6G IA)



Bharat 6G Alliance (B6GA) and the 6G Smart Networks and Services Industry Association (6G-IA) formalized their partnership through a Memorandum of Understanding (MOU) on 22nd March 2024. This partnership facilitates joint research initiatives, contributions to global standards, and participation in regulatory discussions on future frequency bands.

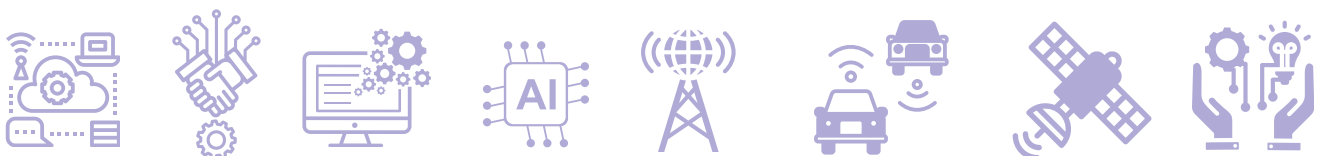
The collaboration will focus on developing innovative solutions, prototyping, and testing, which will benefit from 6G-IA's established European networks and B6GA's Indian expertise. The partnership supports the development of a robust 6G ecosystem in India and Europe.

India's Bharat 6G Alliance to partner with Europe's 6G-IA – report

India's Bharat 6G Alliance is likely to partner with Europe's 6G Smart Networks and Services Industry Association (6G-IA) to jointly develop 6G technology, according to media reports, with an agreement expected to be signed in the coming quarter.

News comes after the Bharat 6G Alliance signed a memorandum of understanding with America's Next G Alliance to work on the development of 6G wireless technologies and to build joint supply chains. These agreements reflect India's ambition of growing its profile as an innovator and a leader in the telecom industry.

The Department of Telecommunications (DoT) launched the Bharat 6G Alliance in August 2023 to boost research and development in 6G. It was formed with the vision to "design, develop and deploy 6G network technologies that provide ubiquitous, intelligent and secure connectivity for a high-quality living experience for the world."



TECHNICAL ACTIVITIES

S. No	Activity	Title	Details
1.	Whitepaper	Potential of OTDFM Waveform & Structural MIMO Technologies	It explores the potential of OTDFM waveform and Structural MIMO technologies as key components of 6G, within the framework of IMT-2030, emphasizing inclusivity, ubiquitous connectivity, sustainability, innovation, security, resilience, standardization, interoperability, and interworking
2.	Bharat 6G Mission Task Force Report	Multi-Platform Next Generation Networks	This report examines global trends in telecom network evolution for 2030 and beyond, identifying and emphasizing the most relevant and promising technological pathways for India's future needs and growth trajectory over the next decade
		Multi-Disciplinary Innovative Solutions	This report explores the transformative potential of 6G technology, highlighting its capabilities of achieving 1Tbps bandwidth, sub-millisecond latency, and integrated sensing services, which are poised to revolutionize human, machine, data interactions across various sectors and discusses its diverse applications in healthcare, agriculture, defence, and other sectors
		6G Spectrum	This report emphasizes the crucial role of efficient spectrum management in accommodating the growing demand for radio-frequency spectrum, driven by society's increasing use of radio-based technologies and the opportunities they offer for social development
		6G Devices	This report highlights the advancements in 6G technology, emphasizing its role in enhancing communication, sensing, imaging, and location awareness, while highlighting the integration of AI-driven decisions in data management and the pervasive presence of future networks in daily life, society, and industries for a sustainable and human-friendly future
		International Standards Contribution	This report discusses how 6G technologies are expected to become impactful over the next decade, enabling ubiquitous communication, pervasive intelligence, immersive experiences, and the Internet of Things & Senses, while supporting societal evolution towards the 2030s and contributing to global sustainability goals, including India's climate objectives
		R & D Finance	This report explores the substantial funding and lengthy R&D cycles required for telecommunication technology products, including the various stages from ideation to commercialization, and highlights how standardized open interfaces have allowed Tech startups to emerge and compete in both public and private networks despite the high barriers to entry.

S. No	Activity	Title	Details
3.	Meetings	The 1st Meeting of the Governing Council of Bharat 6G Alliance	Date: 24 May 2023 Chairman: Mr. N. G. Subramaniam
		The 2nd Meeting of the Governing Council of Bharat 6G Alliance	Date: 26 August 2023 Chairman: Mr. N. G. Subramaniam
		The 3rd Meeting of the Governing Council of Bharat 6G Alliance	Date: 5 February 2024 Chairman: Mr. N. G. Subramaniam
		The 1st Annual General Meeting (AGM) was held on 5th February 2024.	Date: 5 February 2024
		The 1st Meeting of Green and Sustainability Working group of Bharat 6G Alliance	Date: 19 March 2024 Chairman: Mr. Sachin Gupta
		Meeting with NextG Alliance to explore collaboration	Date: 22nd March 2024 Chairman: Mr. N. G. Subramaniam
		Signing of MOU with 6G SNS IA	Date: 22nd March 2024 Chairman: Mr. N. G. Subramaniam
4.	Conference Engagement	Key Drivers for 6G: Use Cases, Technologies, and Other Imperatives	Date: 14th November 2023 Attendee: (i) Dr. Kumar Sivarajan, Co-founder and Chief Technology Officer of Tejas Networks, India (ii) Mr. Ramesh Singh, Chairman, VVDN Technologies (iii) Mr. Manish Mangal, Global Business Head, 5G & Network Services, Tech Mahindra (iv) Mr. Rajesh Kumar Pathak , Member Secretary, Bharat 6G Alliance
		16th International Conference on COMmunication Systems & NETworkS (COMSNETS)	Date: 3-7 January 2024 (Bangalore) Attendee: Mr. Rajesh Kumar Pathak
		Beyond 5G International Conference, Japan	Date: 1-2 February 2024 Attendee: Mr. Rajesh Kumar Pathak

FINANCIAL RESULTS

BHARAT 6G ALLIANCE
BALANCE SHEET AS AT MARCH, 31, 2024

PARTICULARS	NOTE	AS AT 31.03.2024
I SOURCES OF FUNDS		
1 GENERAL FUNDS		
Trust Fund	1	-
Reserves & Surplus	2	(16,460)
2 NON - CURRENT LIABILITIES		
Secured Loans	3	-
Unsecured Loans	4	-
3 CURRENT LIABILITIES		
Other Current Liabilities	5	40,38,000
TOTAL		40,21,540
II APPLICATION OF FUNDS		
1 NON CURRENT ASSETS		
FIXED ASSETS		
Gross Block Tangible Assets	6	5,83,585
Less : Depreciation Fund		1,16,717
Net Block Tangible Assets	7	4,66,868
2 CURRENT ASSETS		
Sundry Debtors		14,24,000
Cash & Bank Balances	8	19,07,477
Short Term Loans & Advances	9	2,23,195
TOTAL		35,54,672
		40,21,540

For Prem P. Lamba & Co.
Chartered Accountant
(Prem Prakash Lamba)
Partner
Membership No: 096654
Firm Regn no: 004244N

For and on Behalf of board of M/s
BHARAT 6G ALLIANCE

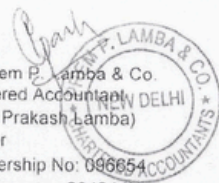
Rajesh Kumar Pathak
(Member Secretary)



BHARAT 6G ALLIANCE
STATEMENT OF INCOME & EXPENDITURE FOR THE YEAR ENDING MARCH 31, 2024

PARTICULARS	NOTE	Period Ended 31.03.2024
I INCOME		
Membership Fess	9	400000
Interest Received	10	11,847
TOTAL		4,11,847
II EXPENDITURE		
Administrative Expenses	11	3,11,015
Financial Expenses	12	575
Depreciation	6	1,16,717
TOTAL		4,28,307
Surplus/ Deficit		(16,460)
Add: Surplus/ Deficit of Previous Year		-
Balance Transferred to Balance Sheet		(16,460)

For Prem P. Lamba & Co.
Chartered Accountants
(Prem Prakash Lamba)
Partner
Membership No: 096654
Firm Regn no: 004244N



For and on Behalf of board of M/s BHARAT 6G ALLIANCE

Rajesh Kumar Pathak
(Member Secretary)



BHARAT 6G ALLIANCE
NOTE FORMING PART OF THE BALANCE SHEET

NOTE:1 TRUST FUNDOpening Balance

Add: Corpus for the year

TOTAL

NOTE:2 RESERVE & SURPLUSOpening Balance

Add Surplus/ Deficit of Current Year

TOTAL

(16,459.99)

(16,459.99)

NOTE:5 CURRENT LIABILITIES

Sundry Creditors

10,000.00

TDS Payable

21,000.00

Advance From Customers

1,77,000.00

Membership Fees

38,30,000.00

TOTAL

40,38,000.00

NOTE:8 CASH & BANK BALANCES

Balance with Bank

18,96,567.00

Cash In Hand

10,910.00

TOTAL

19,07,477.00

NOTE FORMING PART OF THE BALANCE SHEET

NOTE:9 LOANS & ADVANCES

TDS Receivable

1,99,000.00

GST Receivable

24,195.26

TOTAL

2,23,195.26

NOTE:10 INCOME

membership Fees

4,00,000.00

Interest on Saving Bank

11,847.00

TOTAL

4,11,847.00

NOTE:11 ADMINISTRATIVE EXPENSES

Audit Fee

10,000.00

Consultancy Charges

2,10,000.00

Conveyance

1,141.00

Travelling Expenses

44,117.00

Travelling Expenses Foreign

29,262.00

Staff Welfare

749.00

Printing & Stationary

2,200.00

Salary

13,548.00

Short & Excess

(2.01)

TOTAL

3,11,014.99

NOTE:12 FINANCIAL EXPENSES

Bank Charges

575.00

TOTAL

575.00

SUNDRY CREDITORS

Prem P Lamba & Co.

10,000.00

TOTAL

10,000.00

Rajesh Kumar Pathak



[Signature]



BHARAT 6G ALLIANCE
FIXED ASSETS AS ON 31.03.2024

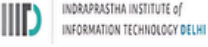
Note 6

	GROSS BLOCK				TOTAL AS ON 31.03.2024	RATE OF DEP.	UP TO 31.3.2023	Adjustment	DEPRECIATION		NET BOOK VALUE	
	COST AS ON 01.04.2023	ADDITION		SALE/TRF					FOR THE PERIOD	TOTAL DEP UPTO 31.03.2024	WDV AS ON 31.03.2024	WDV AS ON 31.3.2023
		BEFORE SEP.	AFTER SEP.									
fixtures	-	-	3,58,584.75	-	3,58,584.75	40%	-	-	71,717.00	71,717.00	2,86,867.75	-
present	-	-	2,25,000.00	-	2,25,000.00	40%	-	-	45,000.00	45,000.00	1,80,000.00	-
	-	-	-	-	-	-	-	-	-	-	-	-
	0	-	-	-	-	-	-	-	-	-	-	-
Total	-	-	5,83,584.75	-	5,83,584.75	-	-	-	1,16,717.00	1,16,717.00	4,66,867.75	-
	-	-	-	-	-	-	-	-	-	-	-	-
TOTAL	-	-	5,83,584.75	-	5,83,584.75	-	-	-	1,16,717.00	1,16,717.00	4,66,867.75	-

Rajesh Kumar Pathak
(Member Secretary)



MEMBERS

ANNEXURE

GOVERNING COUNCIL AS ON 31ST MARCH 2024



**Mr N. G.
Subramaniam**
(Chairman, Bharat 6G
Alliance), TCS



Prof Bhaskar Ramamurthi
(IIT Madras)



Prof Ajit Kumar Chaturvedi
(IIT Kanpur)



Mr Randeep Singh
(Bharti Airtel Ltd)



Mr Aayush Bhatnagar
(Reliance Jio)



Mr Jagbir Singh
(Vodafone Idea Limited)





Dr Rajkumar Upadhyay
(C DoT)



Dr. P Hanumantha Rao
(SAMEER, IIT Bombay)



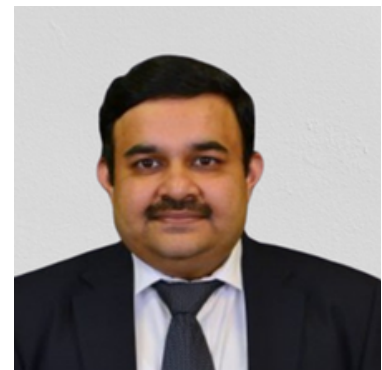
Prof Kiran Kuchi
(Wisig Networks Private Limited)



Mr Ramu T S
(Lekha Wireless Solutions Pvt Ltd)



Dr. Kumar N. Sivarajan
(Tejas Networks Ltd)



Mr Himamshu Khasnis
(Signalchip Innovations Pvt. Ltd)





Mr Puneet Agarwal
(VVDN)



Mr Raghuveer
(Nivetti Systems Private
Limited)



Mr Jagdish Mitra
(Tech Mahindra)



Mr Rajesh Kumar Pathak
(Bharat 6G Alliance)



Shri A Robert Jerard Ravi
Department of
Telecommunications (DoT)



WORKING GROUPS

SPECTRUM



Prof David Koil Pillai
(IIT Madras)



Mr. Saurabh Mittal
(Bharti Airtel)



Mr. A k Tiwari
(Reliance Jio)



Mr. Ravi Lakhotia
(Vodafone Idea)

DEVICES, SENSORS AND MANUFACTURING ECOSYSTEMS



Prof Bhaskar Ramamurthi
(IIT Madras)



Dr. P Hanumantha Rao
(SAMEER, IIT Bombay)



Mr. Himanshu Khasnis
(Signalchip Innovations)



TECHNOLOGY



Prof Kiran Kuchi
(Wisig Networks Pvt. Ltd)



Mr. Puneet Agarwal
(VVDN Technologies)



Mr. Jishnu Aravindakshan
(Tejas Networks)



Mr. Ravi Lakhotia
(Vodafone Idea)

APPLICATIONS



Dr. Ravi Gandhi
(Reliance Jio)



Mr. Raghuveer B K
(Nivetti Systems)



Prof Rajesh Sundaresan
(IISc, Bengaluru)



GREEN AND SUSTAINABILITY



Mr. Sachin Gupta
(Bharti Airtel)



Dr. Venkata Subramanian
Viraraghavan (TCS)

OUTREACH



Dr Rajkumar Upadhyay
(C DoT)



Mr Balaji Rangaswamy
(Sooktha Consulting)

6G USECASES AND REVENUE STREAM



Mr. Rajesh Singh
(Vodafone Idea)



Prof. Rohit Budhiraja
(IIT Kanpur)

Bharat 6G Alliance

C DoT Campus (2nd floor),
Mandi Road, Mehrauli,
New Delhi- 110030, India.

